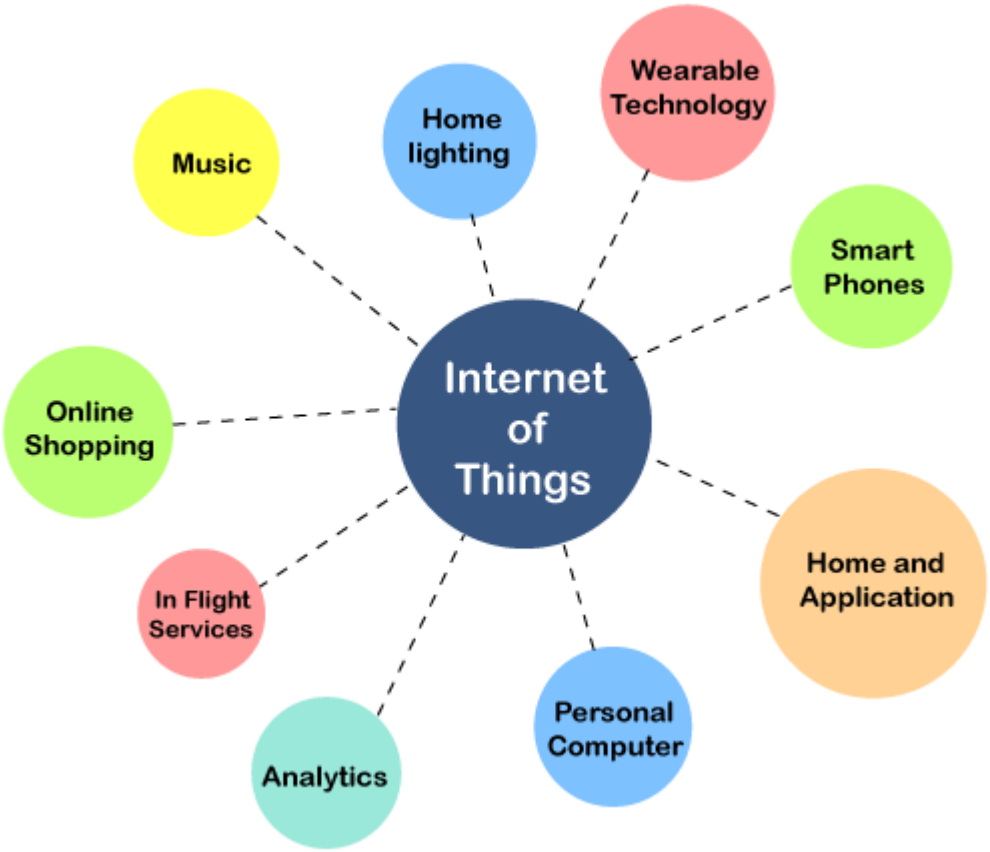
Unit – V: Applications of IoT

5.1 Overview, Block Diagram and Working of the following real world IoT applications

The **Internet of Things (IoT)** provides the ability to interconnect computing devices, mechanical machines, objects, animals or unique identifiers and people to transfer data across a network without the need for human-to-human or human-to-computer is a system of conversation. **IoT applications** bring a lot of value in our lives. The Internet of Things provides objects, **computing devices**, or **unique identifiers** and people's ability to transfer data across a network without the **human-to-human** or **human-to-computer interaction**.



1. **Smart Home Automation:** IoT enables the automation and control of household devices and systems like lighting, thermostats, security cameras, and appliances, providing convenience and energy efficiency.
2. **Industrial IOT:** IoT is used for monitoring equipment health, predictive maintenance and check production processes to increase efficiency and reduced downtime.
3. **Heathcare:** IoT is used for monitoring patient, smart medical devices, improving patient care and timely interventions.
4. **Smart cities:** IoT is used to create intelligent urban environments by monitoring and managing city infrastructure, traffic flow, waste management and public services etc.
5. **Agriculture:** IoT devices like sensors and drones are used in precision farming to monitor soil condition, crop health, weather patterns etc.
6. **Retail:** IoT is used for supply chain, inventory management, customer shopping experience with technologies like smart shelves and personalized marketing.

7. **Transportation and Logistics:** IoT is used to track and optimize transportation routes, monitoring vehicle health etc.
8. **Energy Management:** It is used for smart grid systems, monitor and optimize energy consumption, renewable energy sources.
9. **Environmental Monitoring:** It helps in monitoring and collecting data on environment factors like air and water quality, conservation efforts.
10. **Wearable Devices:** Wearable technology is the hallmark of IoT applications and we have fitness trackers, smartwatches to monitor health metrics and provide personalized insights.
11. **Supply Chain Management:** It is used for real-time tracking of goods, reducing delays and improving overall supply chain.

5.1.1 Smart Home automation

Overview:

- Home automation is constructing automation for a domestic, mentioned as a sensible home or smart house. In the IoT home automation ecosystem, you can control your devices like light, fan, TV, etc.
- A domestic automation system can monitor and/or manage home attributes adore lighting, climate, enjoyment systems, and appliances. It is very helpful to control your home devices.
- It's going to in addition incorporates domestic security such as access management and alarm systems. Once it coupled with the internet, domestic gadgets are a very important constituent of the Internet of Things.
- A domestic automation system usually connects controlled devices to a central hub or gateway.
- The program for control of the system makes use of both wall-mounted terminals, tablet or desktop computers, a smartphone application, or an online interface that may even be approachable off-site through the Internet.
- Smart Home automation refers to the use of technology to control and automate various functions in a home, such as lighting, heating, air conditioning, and security. In the context of IoT (Internet of Things) and M2M (Machine-to-Machine) communications, home automation systems can be controlled and monitored remotely through a network connection.
- One of the key benefits of IoT-enabled home automation is the ability to remotely control and monitor a wide range of devices and systems from a single, centralized location, such as a smartphone or tablet.
- IoT-enabled home automation systems typically involve the use of smart devices, such as thermostats, light bulbs, and security cameras, that can be controlled and monitored through a centralized hub or app. These smart devices can communicate with each other and with the centralized hub using wireless protocols such as Zigbee, Z-Wave, and Bluetooth.

Some key components:

Smart Lighting –

- Smart lighting for home helps in saving energy by adapting the life to the ambient condition and switching on/off or dimming the light when needed.
- Smart lighting solutions for homes achieve energy saving by sensing the human movements and their environments and controlling the lights accordingly.

Smart Appliances –

- Smart appliances with the management are here and also provide status information to the users remotely.
- Smart washer/dryer can be controlled remotely and notify when the washing and drying are complete.

- Smart refrigerators can keep track of the item store and send updates to the users when an item is low on stock.

Intrusion Detection –

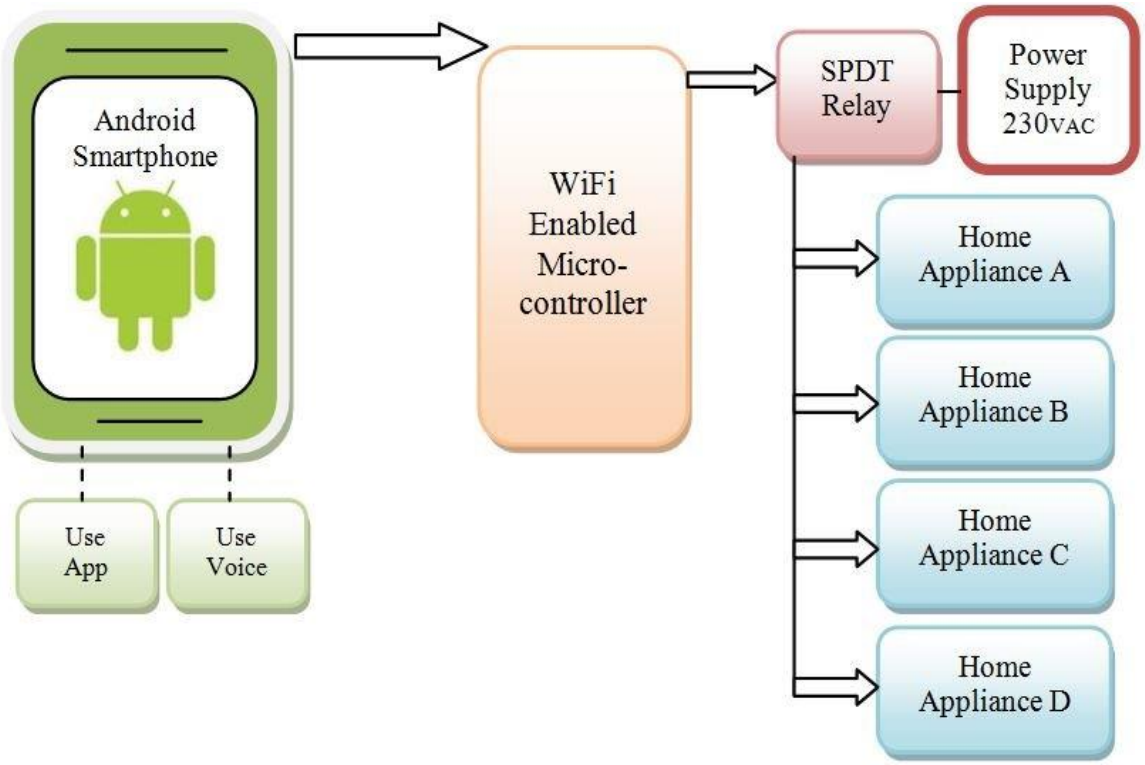
- Home intrusion detection systems use security cameras and sensors to detect intrusion and raise alerts.
- Alert can we inform of an SMS or an email sent to the user.
- Advanced systems can even send detailed alerts such as an image shoot or short video clips.

Smoke/gas detectors –

- Smoke detectors are installed in homes and buildings to detect smoke that is typically an early sign of Fire.
- It uses optical detection, ionization for Air sampling techniques to detect smoke.
- Gas detectors can detect the presence of harmful gases such as CO, LPG, etc.
- It can raise alerts in the human voice describing where the problem is.

Working of Smart Home Automation System:

- For a smart home automation system, we need to create a remotely controllable network.
- All devices like lights, thermostats, security system and appliances are controlled by a master home automation controller. It is known as smart home hub.
- This hub is hardware device that acts as central point. It sense, process data and communicate wirelessly.
- It combines all the disparate apps into a single smart home app that owner can control remotely.
- Example: Amazon Echo , Google Home
- Many smart home system uses Wifi and Bluetooth to connect, some depends on wireless protocols like Zigbee or Z-Wave.
- Some devices are scheduled to commands like Amazon Alexa or Google Assistant.



- Requirements:
  - Home owner have strong internet connection as all relies on internet.
  - Select proper hub to manage central control.
  - Start with basic items like smart plugs or bulbs.
  - For security set up strong passwords and multifactor authentication.
  - Homemaker can add more devices.
- Examples of smart home automation technologies:
  - Smart TVs
  - Smart lighting system
  - Smart thermostats
  - Smart door locks and garage door openers
  - Smart security cameras
  - Smart pet and lawn care
  - Smart Kitchen appliances
  - Smart household monitors
  - Smart plugs

Advantages of Smart Home Automation:

1. Provide assurance: Homemaker can control and monitor remotely, like if coffee maker left on.
2. Accommodates user preferences for convenience: user can program their grarage.
3. Offers peach of mind: Manage health and well-being of seniors remotely.
4. Improves efficiency: manage AC for cooling instead of keep ON all day.
5. Saves resources and money: Like with irrigation we can save water.
6. Manage tasks: Homemakers cam manage task with voice commands.

Disadvantages of Smart Home Automation:

1. Require a reliable internet connection
2. Complexity of using technologies
3. Lack of Standards
4. Questionable security
5. Lack of data privacy
6. Expense

5.1.2 Agricultural System

Overview:

IoT based smart agriculture integrates various technologies and sensors to manage agriculture processes, leading to increased efficiency and resource optimization.

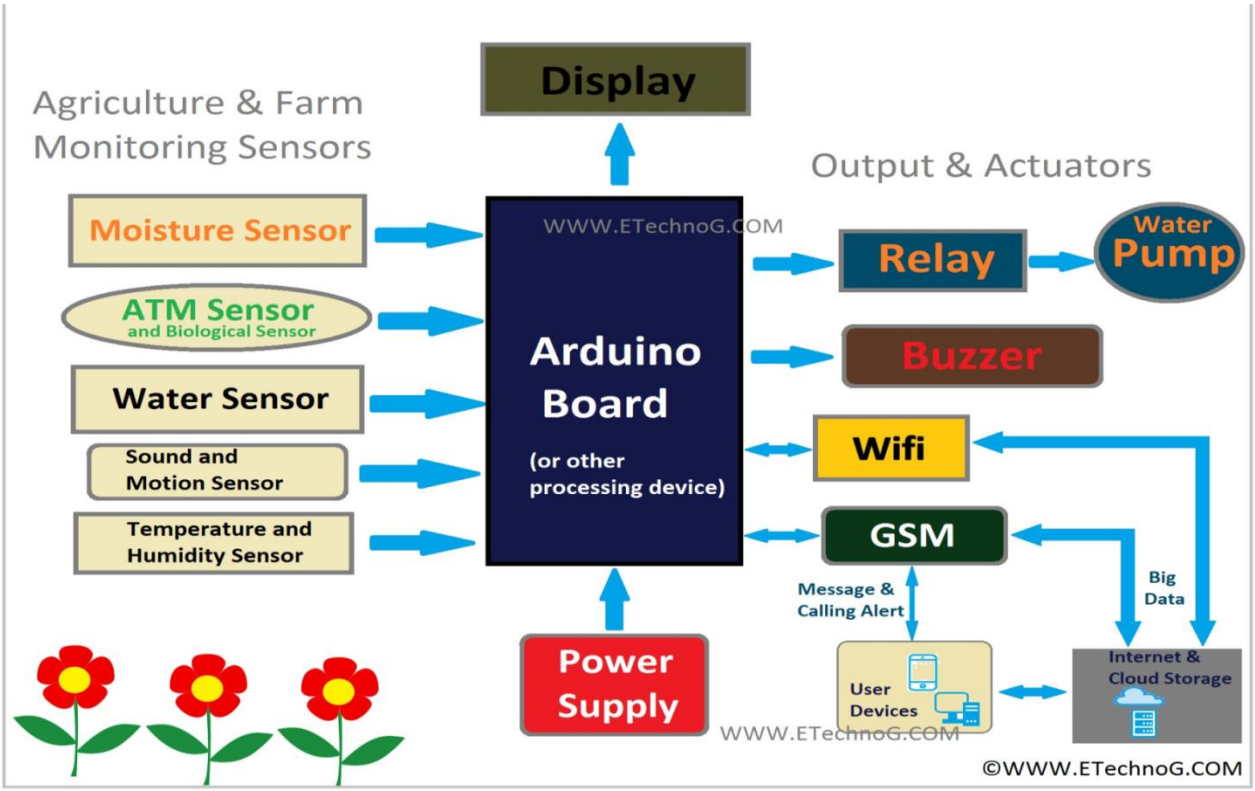
Sensors Deployment/ components:

- **Soil sensors:** These sensors measure soil moisture, temperature, nutrient levels. They can measure temperature, humidity, wind speed, helping farmers for crop management.
- **Wireless connectivity:** Sensors are connected with wireless communication (Wi-Fi, Zigbee) to transmit data to central hub or cloud platform.

- **Centralized cloud Platform:** Cloud services help store, process and analyze agricultural data.
- **Data Analysis and Decision Support:** Advanced analytics processes the collected data to provide insights into crop health and growth pattern.
- **Remote monitoring and control:** Farmers can monitor their crops and field conditions using web-based or mobile applications.
- **Automated Irrigation:** Crop receives the right amount of water using automated irrigation system.
- **Crop health monitoring:** Using drones with cameras or imaging sensors we can capture visual data about crop health. So we can easily identify diseases or deficiencies.
- **Livestock monitoring:** Wearable sensors on animals can provide data on their well-being. Also monitor location of livestock.

Working of Agricultural System:

- Arduino board is main part of the whole system.
- It takes all the signals, data and information through different types of sensors.
- We can connect our smartphones through internet. We can see all the status of our farm in mobile app.
- Ex: check water level, soil moisture level etc.
- Water level sensor will send data to Arduino and Arduino will send it to our phone.
- If the water level is low then we give a command to start water pump from smartphone.
- So Arduino will receive command and start pump by activating relay switch.
- Again when water is filled water sensor send signal to Arduino, it will send status to phone.
- Now we can give the command to stop water pump, like same process, it will stop the pump.



Advantages of Agricultural System:

1. **Precision Farming:** By providing real-time data on soil conditions, weather patterns and crop health we can do precision farming.
2. **Resource Efficiency:** This leads to more efficient use of resources, reducing waste and lowering production costs.
3. **Remote monitoring control:** Monitoring livestock, receiving real-time updates on field conditions and animal health.
4. **Optimized crop management:** It helps to identify diseases, accessing nutrient levels.
5. **Cost savings:** Reducing waste, resource usage and overall efficiency maintains so cost savings for farmers.
6. **Environmental sustainability:** Smart agriculture promotes sustainable farming practices by minimizing environmental impact of farming activities.
7. **Market Access:** It improves farm's ability to meet regulatory and market requirements.

Disadvantages of Agricultural System:

1. **Technical Complexity:** It involves different technologies, including sensors, communication network and data analytics.
2. **Data security and privacy:** Farmers may be hesitating to share sensitive information. There is need of robust data protection.
3. **Dependency on connectivity:** Smart agriculture heavily relies on stable and reliable internet connectivity. In rural area there is limited access to high speed internet.
4. **Power Supply Dependency:** Automated system required stable power supply, So farmers may face challenges.
5. **Limited customization for small farms:** Smart agriculture solution is may be designed for large commercial farms, less suitable for small-scale farmers.
6. **Environmental Impact:** Manufacturing and disposal of electronic devices and components used in smart agriculture technology, if it cannot manage properly then contribute to electronic waste.
7. **Possibilities of cyber attack:** Due to internet usage, there is possibilities of cyber attack which leads to data loss, cyber threats, risk to physical infrastructure.

5.1.3 Smart Parking

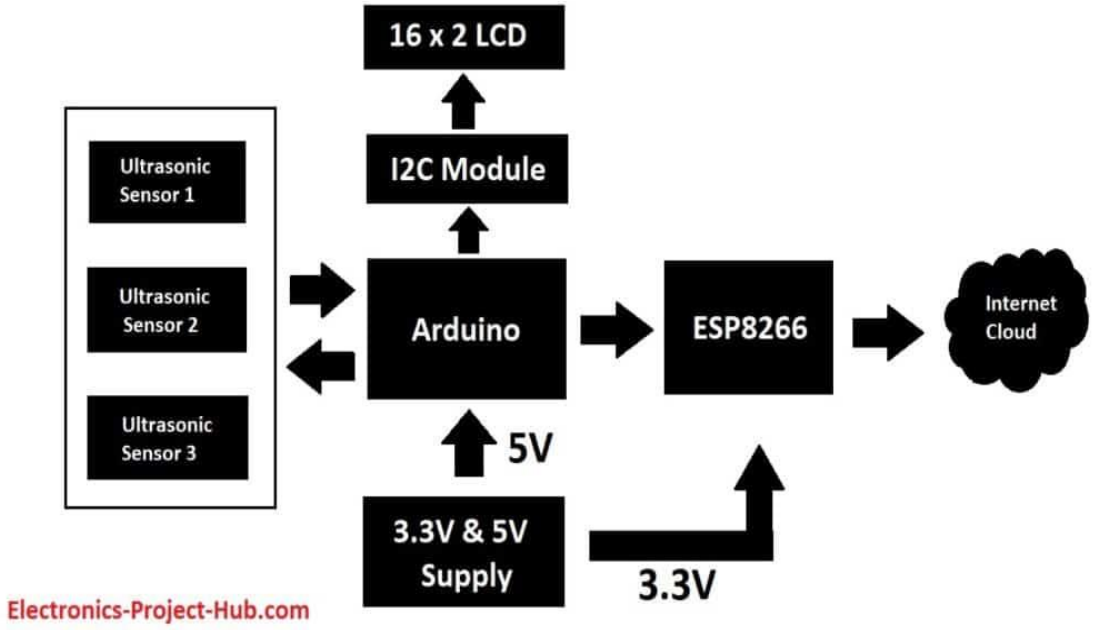
Overview:

- IoT based parking system is vehicle parking management system.
- It is used to search a free/vacant parking spot in the parking area using smartphone.
- It is used to manage parking spaces efficiently.

Working of Smart Parking:

- **Sensor Deployment:** These sensors can detect the presence or absence of vehicles and communicate information wirelessly.
- **Wireless Connectivity:** To transmit real-time data to a centralized system.
- **Centralized System/Cloud Platform:** Data is sent to cloud for processing and analysis. Cloud platforms use data analytics algorithms to interpret information and generate insights about parking space availability.

- **Mobile Application and User Interface:** Mobile apps display real-time updates, maps and navigation features to guide drivers for nearest parking.
- **Reservation and Payment Features:** It allows users to reserve parking spaces in advance.
- **Integration with Navigation Systems:** It allows drivers to receive real-time parking information as part of their route planning.
- **Occupancy monitoring and Historical data:** System monitors free parking space, providing administrators with real-time historical data.
- **Alert and Notifications:** Users receive alerts and notifications about their parking sessions, upcoming expirations.
- **Example:** Figure given below is one example of smart parking system:



1. A 16 * 2 LCD is used for displaying no of free sports
2. An I2C module is used for driving LCD.
3. There are 3 ultrasonic sensors for detecting 3 vehicles on parking spot.
4. Ultrasonic sensor acts like a mini radar and environmental factors affecting its functionality is minimal.
5. ESP8266 Wi-Fi module is used for internet connectivity which sends the free slot data to cloud server for view data in real time.
6. Power supply module is used for 5V and 3.3V for Arduino, ultrasonic sensors and Wifi module.
7. Cloud service here used is known as “Thingspeak” where the parking lot’s data to be sent, stored and displayed in real time.

Advantages of Agricultural System:

1. **Improved Parking Efficiency:** Helping drivers to find parking quickly and reduce searching time.
2. **Reduce Traffic Congestion:** As we get quick parking, it reduces congestion specially in urban areas.
3. **Space Utilization:** Provides better utilization of available resources.
4. **Enhanced user experience:** Drivers get stress free parking experience.
5. **Cost Savings for Drivers:** Drivers can save on fuel cost by minimizing the time for parking.
6. **Integration with Navigation Systems:** Drivers get real time parking information for route planning.
7. **Enhanced Safety and Security:** Smart have surveillance cameras and emergency call buttons for safety.

Disadvantages of Agricultural System:

1. **High Initial Costs:** Sensors, communication networks, central system, all involve more costs.
2. **Dependency on Technology:** It rely heavily on technology, so system failures, connection issues can disturb the functionality.
3. **Complexity and Learning Curve:** Drivers and parking facility operators may need to understand the use of technology.
4. **Power supply Dependency:** All devices like sensors require power. Areas where less electricity can be challenging.
5. **Privacy Concerns:** Data collection, storage and use may arise privacy issues.
6. **Maintenance Complexity:** Complexity of technology and need of specialized skills can make maintenance challenging.
7. **Reduction in jobs for Attendants:** Due to atomization in functions requires less human power so reduction in job opportunities.

VPMP PC